

A Local-First Autonomous Trading Research Desk: Sub-Second Voice Control, and What 7,312 Forward Tests Showed

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Abstract—We describe AXIOM, a research desk that runs entirely on one workstation: it places paper trades across seven strategies, keeps a written record of what it learns, and is operated by speaking to it. Two results are reported. First, a voice loop built from open models (Whisper through `whisper.cpp` for listening and Piper for speaking) answers a spoken question in 0.69 to 1.39 seconds end to end, with transcription at 105 to 215 ms, and needs no network and no third-party service. Reaching that latency required discarding two designs that appear correct and are not: recording microphone audio with `MediaRecorder`, which produces undecodable fragments when an utterance is cut from the middle of a stream, and gating speech against absolute energy thresholds, which on the measured hardware could never open. Second, and more importantly for anyone building such a system, the desk’s own record argues against most of its strategies. Over 61 days and 7,312 paper trades the aggregate profit factor is 0.81 and expectancy is $-\$1.80$ per trade; a Monte Carlo permutation test clears exactly one of seven strategies as robust and marks one marginal. The single consistently profitable book is a weather prediction-market strategy whose edge appears only after an entry gate is applied: the same signal, ungated, lost $\$212.67$ over 248 trades, and gated, returned $+\$88.84$ over 279 trades at an 83.5% hit rate. We argue that the value of a system like this lies less in the strategies it runs than in the discipline of forward-testing them honestly in public view, and we report the negative results in full.

Index Terms—algorithmic trading, forward testing, speech recognition, voice user interfaces, on-device inference, prediction markets, Monte Carlo permutation testing

I. INTRODUCTION

A retail algorithmic trader today has no shortage of tools and a serious shortage of trustworthy evidence. Backtests are cheap, easy to overfit, and almost always flattering; the literature on this is long and consistent [1]–[3]. The honest remedy is forward testing: run the strategy on data it has never seen, in something close to real conditions, and keep the record whether or not it is comfortable reading.

This paper describes a system built around that remedy. AXIOM is a single-user research desk that runs on one

Apple Silicon workstation. It hosts seven strategy daemons that trade on paper against live market data, a reflection loop that revises their parameters and writes down what it concludes, a live globe that carries open-source intelligence feeds, and a voice interface through which the operator asks questions and gives instructions in ordinary speech.

We make three contributions.

A measured local voice loop. Section IV reports a complete speech interface covering wake word, barge-in, transcription, language model and synthesis, running on the operator’s own machine with no audio leaving it. We give the latency budget stage by stage and the accuracy-versus-latency tradeoff across four speech recognition models.

Two negative engineering results that cost real time. Both concern designs that look correct in a code review and fail in use. Browser `MediaRecorder` output cannot be sliced by a voice activity detector, because every chunk after the first lacks a container header and, once a header is restored, carries a timestamp that makes the decoder emit the entire session as silence. Separately, an energy gate with any absolute threshold is untenable across microphones: the device measured here peaks at 0.0016 RMS, so a threshold of 0.004, a value that seems conservatively low, can never open.

A full and unflattering forward-test record. Section VI reports every strategy the desk runs, including the six that do not survive their robustness test and the one that lost $\$11,277$ before it was retired.

II. RELATED WORK

A. Evaluating trading strategies

The difficulty of distinguishing skill from selection in backtested strategies is well documented. White’s reality check [1] and the deflated Sharpe ratio of Bailey and López de Prado [2] both address the multiple-testing problem directly; Harvey and Liu [3] argue that most published factors do not survive an appropriate haircut. Our robustness gate is a simpler device, a permutation test on trade order from which we take a probability that the observed win rate arose by chance, but the motivation is theirs.

B. Open speech models

Whisper [4] established that a single supervised model can transcribe robustly across accents and noise; `whisper.cpp` [5] made it practical on consumer CPUs through quantisation and platform-specific kernels. Piper [6] provides neural speech synthesis at similar cost. Both are permissively licensed, which is what makes an entirely local voice loop possible. Commercial dictation products such as Wispr Flow are closed, but the open alternatives are built on precisely these two components.

C. Open-source intelligence feeds

The globe described in Section III aggregates public transponder, orbital and seismic data from OpenSky [7], CelesTrak [8] and the USGS earthquake feed respectively, in the tradition of open-source intelligence dashboards. We claim no novelty here; it is context for the strategies that consume weather and event data.

III. SYSTEM ARCHITECTURE

The desk is four layers on one machine.

Strategy daemons. Seven independent processes, each supervised by `launchd`, poll their own data sources and write every decision to an append-only JSON-lines log. A daemon never holds state only in memory; the log is the record, and every figure in this paper is derived from those logs rather than from any in-process accounting. We verified this property directly: the weather book’s reported +\$88.84 was recomputed from `dryrun_weather.jsonl` by joining each placed bet to the market’s own published resolution and applying the same fee model, and it reproduced to the cent over the same 279 trades.

Reflection loop. A nightly process reads the logs, fits simple summaries per strategy and per market segment, and writes back a parameter file and a dated note in prose. Its conclusions are visible on the desk’s home screen. It concluded, for example, that one city was disabled in the weather book and that the minimum edge to act was raised to 0.08.

Interface. A Next.js application serves roughly forty pages. Relevant here are the journal, which renders the forward-test record, and the globe, which carries the intelligence feeds.

Voice. A bridge process holds the conversation, calls tools on the operator’s behalf, and streams its answer back sentence by sentence. Speech recognition and synthesis run in a separate always-warm Python service.

IV. THE VOICE INTERFACE

A. Capture

Audio is taken from the microphone through an `AudioWorklet`, so that capture runs on the audio thread and is unaffected by whatever the interface is rendering. We initially used a `ScriptProcessorNode` on the main thread; with the globe and charts animating it dropped

buffers, and sentences arrived at the recogniser chopped into single words.

Frames are held in a ring buffer of 500 ms so that the first word survives the delay before the gate opens, and speech is considered finished after 420 ms of quiet. The utterance is decimated to 16 kHz, normalised, and written as a WAV file in the browser.

B. Why not MediaRecorder

The obvious implementation records with `MediaRecorder` at a fixed timeslice and concatenates the chunks belonging to an utterance. It does not work, for two reasons that only appear at runtime. The first chunk carries the WebM header and every later chunk is a bare cluster, so an utterance taken from the middle of the stream has no header and `ffmpeg` refuses it outright; in our case every spoken command failed with HTTP 503. Prepending the stored header fixes the container but not the content: the cluster retains its original timestamp, so the decoder emits everything since the session began as silence, and the recogniser hallucinates over the gap. We observed [MUSIC PLAYING] and “I’m on camera” returned for a four-word command. Writing PCM directly avoids both problems and costs about twenty lines of code.

C. Voice activity detection

The gate is a two-threshold energy detector with hysteresis, and, in the part we got wrong twice, it is expressed purely as a multiple of the room’s own noise floor, which is tracked continuously, including while the operator is speaking. Our first version tracked the floor only between utterances. Because a real microphone never reaches digital silence, the gate then never closed, and every utterance ran to the fifteen-second ceiling: we measured 467 KB of audio sent for a four-word command.

The second version added an absolute term, $\text{floor} \times 3.5 + 0.004$. On the hardware measured here the microphone peaks at 0.0016 RMS, so that gate could not open at all. An absolute energy threshold is not portable across microphones, gain settings, or how far the speaker sits from the device; we recommend against one.

D. Barge-in and echo

The microphone stays open while the system speaks, so the operator can interrupt mid-sentence, as one would with a person. This means the microphone also hears the loudspeaker. Browser echo cancellation and a raised gate remove most of it; the remainder is removed in software.

Our first echo filter compared the transcript against a bag of words the system had recently spoken and discarded anything overlapping by more than half. This is wrong in a way that is easy to miss and severe in use: the system talks about the same subjects the operator asks about. Tested against realistic follow-ups, it suppressed four of five, including “pause the flow bot”. The system ignored its operator precisely when he used its own vocabulary.

Echo is better characterised as the same words in the same *order*, as one unbroken run. Let h be the transcript and m a recent utterance, both as word sequences, and let $r(h, m)$ be the length of the longest run of consecutive positions matching under a fuzzy word comparison. We reject the transcript when

$$r(h, m) \geq 4 \quad \text{and} \quad \frac{r(h, m)}{|h|} \geq 0.85. \quad (1)$$

Measured against real answers, echo scores 1.00 on this ratio and genuine follow-ups score between 0.29 and 0.75, so the threshold separates them cleanly.

E. Addressing

A wake word starts a conversation but does not punctuate one: for 90 seconds after any exchange, speech is taken as directed at the system, subject to a ten-minute ceiling so that a room full of conversation cannot hold the channel open indefinitely. The wake pattern is deliberately permissive, because the recogniser’s errors are concentrated there. We observed “Hay Axiom”, “Axem”, “Acts him” and “action” for the same name. It also accepts ordinary forms of address such as “hey buddy”.

F. Latency

Fig. 1 gives the budget. The dominant fixed cost is the hold-off that decides speech has ended, which is a design choice rather than a computation. Transcription of a typical utterance takes 105 to 215 ms and synthesis 148 to 360 ms.

Three implementation details account for most of the improvement over our first working version. The browser sends 16 kHz mono PCM, which is what the recogniser wants, so the service passes samples straight to the model rather than spawning `ffmpeg` to convert the audio into the format it was already in; this alone took transcription from 334 ms to 105 ms. Both models are exercised with a real inference at start-up, because loading a model and building its compute graph are different things and the operator was paying for the latter on his first word after every restart. And both run off the event loop under a lock, because each holds the CPU: before this, a sentence being spoken blocked transcription of the next thing said, turning a 200 ms render into eleven seconds.

G. Choice of recognition model

Fig. 2 plots word error against transcription time for four Whisper variants on a fixed set of the desk’s own commands. The quantised `large-v3-turbo` model is perfect on this set and takes 2.17s per utterance; it was CPU-bound and became slower, not faster, with more threads. The `base.en` model takes 148 ms at 5.7% word error.

That comparison is misleading on its own, because the errors are not uniformly distributed. Every error `base.en` made was a spelling of the system’s name. Measured on the remainder of the sentence, the part that determines what the system actually does, `base.en` scores 0.0% word error

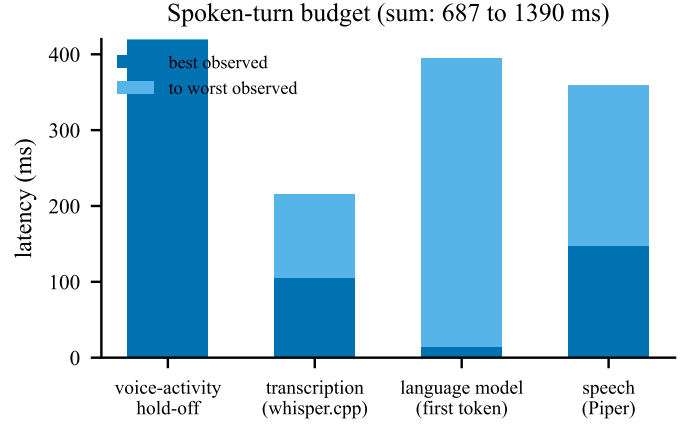


Fig. 1. Measured latency of each stage of a spoken turn. Bars show the best and worst observed values over repeated runs on an otherwise loaded workstation.

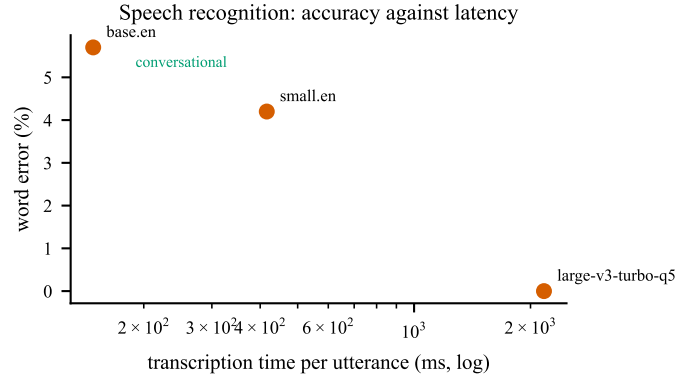


Fig. 2. Accuracy against latency for four Whisper variants on the desk’s own command set. `tiny.en` is omitted from the error axis because it failed to recognise the wake word at all.

over the same set, and correctly recognised that it was being addressed in seven cases out of seven. We therefore kept the small fast model and widened the wake pattern, rather than paying 1.3s per sentence to have the name spelled correctly.

V. FORWARD-TEST METHODOLOGY

Each strategy trades a notional paper account and records every decision. Position sizes are fixed per strategy (\$10 to \$30) so that results are comparable and no single book can dominate through leverage. Fills are taken at the quoted price at decision time; we do not model queue position or partial fills, which is a real limitation discussed in Section VIII.

Fees are charged on every trade. For prediction-market trades we apply the venue’s published formula, which is a function of stake and entry price, $\text{fee} = 0.018 \times 4 \times s \times p(1 - p)$ for stake s at price p . All figures in this paper are after fees. The difference is not cosmetic: the weather book’s gross return of +\$120.38 becomes +\$88.84 once fees are applied, a reduction of 26%.

TABLE I
AGGREGATE FORWARD-TEST RECORD, 61 DAYS, AFTER FEES

Trades	7,312
Win rate	45.2%
Gross profit	\$55,472.47
Gross loss	\$68,606.09
Net	−\$13,133.62
Profit factor	0.81
Expectancy per trade	−\$1.80
Average win / average loss	\$16.78 / \$17.13
Largest win / largest loss	\$1,912 / −\$2,555
Maximum drawdown	\$15,418.50
Longest winning / losing run	21 / 19

Composition of gross loss (total \$13,342)

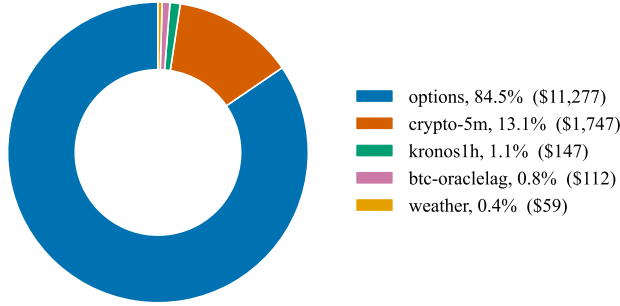


Fig. 3. Composition of gross loss by strategy. One book dominates; the remainder are small by comparison.

Robustness. For each strategy we compare the observed win rate against the distribution obtained by permuting trade outcomes, and report the probability that the observed result arose by chance. We label a strategy robust when that probability is small and its expectancy is positive, marginal when the evidence is weak in either direction, and not robust otherwise. This is a coarse instrument. It does not correct for the number of strategies tried, which by the arguments of [3] it should, and we treat its verdicts as a filter for attention rather than as proof.

VI. RESULTS

A. Aggregate

Table I gives the desk’s whole record. It loses money. Over 61 trading days and 7,312 trades the net result is −\$13,133.62, the profit factor is 0.81, and the expected value of a trade is −\$1.80. The average win (\$16.78) and average loss (\$17.13) are nearly equal, so with a 45.2% win rate the arithmetic is unforgiving.

B. Where the losses are

Fig. 3 decomposes the gross loss. It is not evenly spread: a single options strategy accounts for the large majority of it. That strategy was retired, and the experience is the reason the desk now gates a book on evidence before letting it run unattended.

TABLE II
PER-STRATEGY FORWARD-TEST RECORD, AFTER FEES

Strategy	Trades	Win rate	Net (\$)	Verdict
crypto-5m	5,658	45.4%	−1,747.45	not robust
premarket	200	40.0%	+112.51	robust
weather	527	63.8%	−58.86	not robust
kronos1h	52	38.5%	−146.73	not robust
btc-oraclelag	307	55.7%	−111.60	not robust
vwap-trend	375	23.5%	+95.51	marginal
options	193	22.3%	−11,277.00	not robust

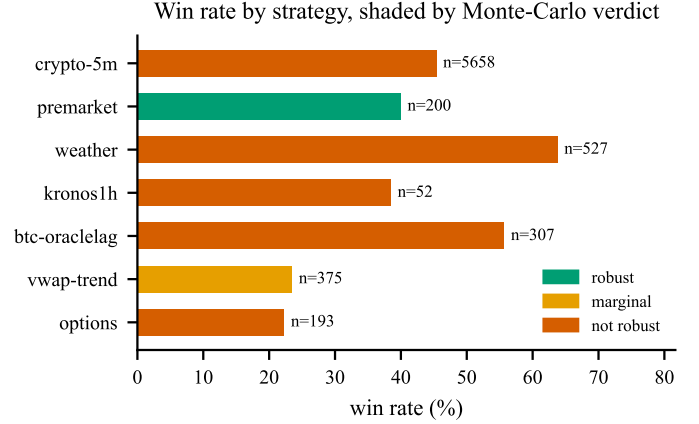


Fig. 4. Win rate by strategy with trade counts, shaded by Monte Carlo verdict. Neither high win rate nor high trade count predicts the verdict.

C. Per strategy

Fig. 4 shows win rate and robustness verdict for each strategy, and Table II the underlying numbers. Of seven strategies, one is robust, one marginal, and five are not. Two observations are worth drawing out.

A high win rate does not imply a positive expectancy. The weather strategy in this table wins 63.8% of its trades and still loses money over the full period, because it includes the ungated era described below. Conversely **vwap-trend** wins under a quarter of its trades and is mildly profitable, because its winners are large relative to its losers.

The strategy with the most trades is the least robust. **crypto-5m** has placed 5,658 trades, 77% of all activity on the desk, at a 45.4% win rate for a loss of \$1,747. Trade count is a measure of activity, not of edge.

D. The one book that works, and why

The clearest positive result concerns the weather strategy, which bets on daily maximum-temperature buckets in prediction markets using station observations and an ensemble forecast. In its first form it traded any bucket whenever its model disagreed with the market. Over 248 trades it won 41.5% and lost \$212.67.

The strategy was then gated: it trades only after hour 14 of the local day, only on favourites priced above 0.75, and

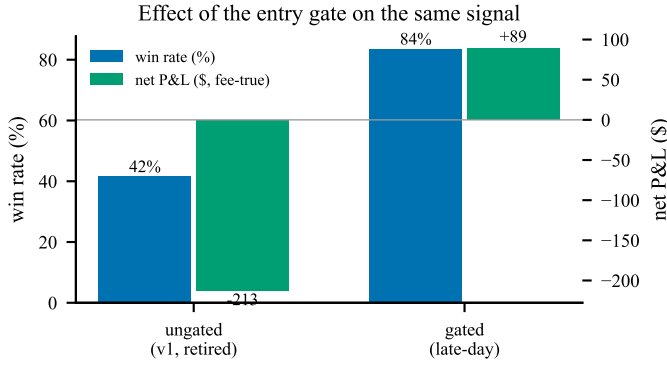


Fig. 5. The same weather signal before and after an entry gate. Win rate and fee-true net P&L over 248 and 279 trades respectively.

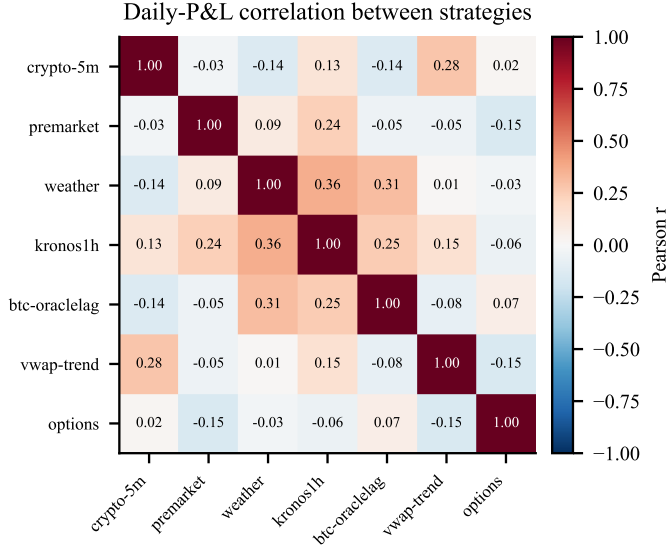


Fig. 6. Pearson correlation of daily P&L between strategies. Low correlation indicates the books are not expressing the same bet.

only when the modelled edge exceeds 0.18. The rationale is that by mid-afternoon the day’s maximum is largely determined by observations already in hand, so the model’s advantage over the market is real rather than speculative. Under the gate, over 279 trades, it wins 83.5% and returns +\$88.84 after fees. Fig. 5 compares the two.

The signal did not change. The decision about when to act on it did, and that is the whole difference between a losing book and a profitable one.

E. Diversification

Fig. 6 shows the correlation of daily P&L between strategies. Most pairs are near zero and several are negative, which is the property one wants: the books fail independently. This is cold comfort while the aggregate expectancy is negative, since uncorrelated losses are still losses, but it means that if any individual book is fixed, its contribution is additive rather than redundant.

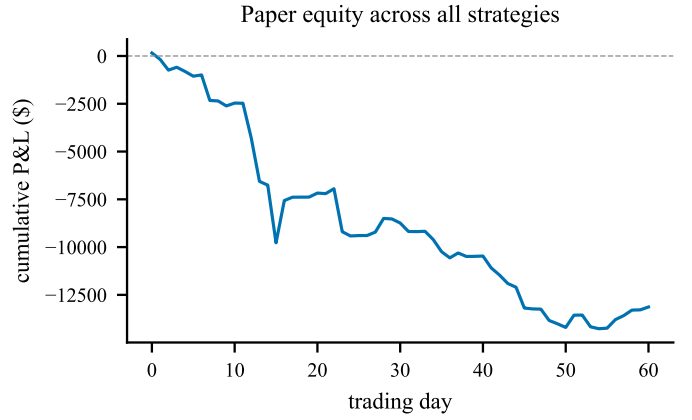


Fig. 7. Cumulative paper P&L across all strategies over the 61-day period.

F. Equity

Fig. 7 shows cumulative paper P&L. The shape is dominated by the options book’s failure early in the period and flattens once it is retired, which is the visual signature of a system that stopped a losing strategy rather than one that found a winning one.

VII. DISCUSSION

The strategies are mostly bad. We think reporting that plainly is the useful contribution, because a reader evaluating a similar system will otherwise see only the one book that works and will draw the wrong conclusion about the base rate.

What the desk demonstrates is not a source of alpha but a method: run the strategy forward on unseen data, charge it real fees, write every decision to a log that can be independently recomputed, and let a robustness test decide where attention goes. Under that method, six of seven candidate strategies were identified as not worth capital, one was found to be salvageable by restricting when it acts, and one modest positive survived. That ratio is consistent with the multiple-testing literature [3] and, we suspect, with most private research that goes unpublished.

The voice interface is a genuine productivity gain, and the reason is latency rather than novelty. Below roughly a second, asking the desk a question is cheaper than navigating to the answer, and the operator asks more questions. A non-trivial amount of the engineering described in Section IV exists to stay under that threshold without sending audio anywhere.

VIII. LIMITATIONS AND THREATS TO VALIDITY

Paper trading is not trading. We assume fills at the quoted price and model neither queue position, partial fills, nor market impact. For the weather strategy, which trades illiquid prediction-market buckets at \$10 a position, this assumption is optimistic in a way we cannot currently bound.

The period is short. Sixty-one days spans one broad market regime. No strategy here has been tested across a regime change.

The robustness test is coarse. A permutation test on trade order does not correct for the number of strategies tried, and by the standards of [2] our “robust” label is weaker than it sounds. The 200-trade sample behind that label is small.

Single operator, single machine. All speech measurements come from one speaker, one microphone and one workstation. The microphone level that broke our energy gate is one data point; we report it because an absolute threshold failed, not because 0.0016 RMS is representative.

Venue access is a live constraint. The only consistently profitable strategy trades a venue that is not legally accessible from the operator’s jurisdiction, which is an unresolved practical problem and not merely an administrative one.

IX. CONCLUSION

We built a research desk that runs on one machine, speaks and listens without a network, and keeps an auditable record of every trade it places. The voice loop answers in about a second, and getting there was mostly a matter of discarding two plausible designs, container-based audio capture and absolute energy gating, that fail for reasons invisible until they are measured.

The trading record is negative in aggregate and we report it in full: 7,312 trades, a profit factor of 0.81, and one strategy in seven surviving its robustness test. The one book that does work became profitable not through a better model but through a restriction on when it is allowed to act. We think that is the most transferable finding here, and we would rather publish it alongside the losses than on its own.

AVAILABILITY

The desk, including the strategy daemons, the voice service and the analysis used for every figure in this paper, is maintained as a single repository. All figures are generated directly from the live logs by the script accompanying this paper; no values were transcribed by hand.

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